

Module 2: The Atmosphere

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1 Overview

While the atmosphere has a pretty constant pressure and composition (globally and on average) at the surface of the Earth, it has incredible variation in temperature, pressure, and composition with altitude, latitude, and longitude. It also varies with solar activity, responding to the solar cycle and solar impulsive events. Winds, tides, and gravity waves further complicate the atmospheric density and flow velocity. What's more, the density and composition are particularly difficult to measure (why?!).

The Earth's atmosphere is the planet's main line of defense against radiation from the sun. It absorbs solar UV and X-rays, as well as energetic particles incident from space. But for spacecraft in low Earth orbit, it also imposes a drag force that limits the lifetime of spacecraft at low altitudes (of course, shortens the orbital lifetimes of space debris as well).

2 Key Points

- The Earth's atmospheric density decreases exponentially with altitude up to over 1000 km
- The atmospheric temperature profile goes through a number of maxima and minima
- The atmosphere temperature, pressure, and composition vary with altitude, latitude, longitude, and solar input, and are subject to winds, tides, and other variations
- The atmospheric density is high enough to affect satellite orbits up to at least 600 km altitude
- Atmospheric atoms and molecules have other problematic effects on spacecraft

Reading:

- Required: Pisacane, Chapter 12; bits of Chapter 14
- Optional: Tribble, Chapter 3

3 Characterization and Theory

The Earth's atmosphere decreases exponentially with altitude starting from the Earth's surface. At altitudes where most of our spacecraft operate – in Low Earth Orbit, or LEO – the density is extremely low compared to the surface, but there is still enough atmosphere to have significant effects on spacecraft. In fact, the atmosphere is the **primary source** of uncertainty in spacecraft and space debris orbit prediction.

3.1 Regions of the Atmosphere, by Composition

Atmospheric layers occur due to different solar absorption coefficients for different constituents, differing masses of constituents, circulation and convection effects, and waves and tides. To start, let's discuss the atmospheric regions by the composition of constituents that populate them. The atmosphere down here on

Constituent	Formula	Molecular Mass	% by Volume
Nitrogen	N ₂	28.0134	78.1
Oxygen	O ₂	31.9988	20.9
Argon	Ar	39.948	0.93
Neon	Ne	20.183	0.00182
Helium	He	4.0026	0.000524
Methane	CH ₄	16.04303	0.00020
Krypton	Kr	83.80	0.000114
Hydrogen	H	2.01594	0.00005
Xenon	Xe	131.30	0.0000084

Table 1: Composition in the troposphere (i.e. near the Earth's surface).

the surface is primarily N₂ (78%), O₂ (21%), and Argon (~1%), plus a large number of minor species. CO₂, for example, is globally about 400 parts per million by volume (ppmv, or about 0.04%), while helium is only 5 ppmv. Water vapor, the source of humidity, can add an additional 0.001 – 7% to the atmosphere by volume.

The atmospheric density decreases exponentially with altitude, with a scale height of about ~8 km; that means every 8 km the density decreases by a factor of $e = 2.718$. This can be written as

$$\rho = \rho_0 e^{-z/H} \quad (1)$$

where z is the altitude in km, H is the scale height¹, and ρ_0 is the mass density at the surface, about 1.3 kg/m³.

This scaling law holds true, and the composition of the atmosphere has the same fractions, up to an altitude of about 100 km. This region from the ground to 100 km is known as the *homosphere*, since the atmosphere is homogeneous. Above 100 km, the atmospheric composition begins to change. The solar radiation impinging on the atmosphere from above causes photodissociation² of N₂ and O₂ into N and O. Helium and hydrogen rise through the atmosphere simply because they are lighter than the other gases, and they then settle in the altitudes shown in Figure 1. You can see from this figure that the different atomic constituents organize themselves from heaviest to lightest as we increase in altitude. Above ~100 km, the atmosphere is dominated by O; above 500 km, He becomes dominant; and at the very top of the atmosphere above 1200 km (not shown on this plot), atomic hydrogen becomes the dominant species. The region above 100 km where the composition changes is called the *heterosphere*, and the dividing line is called the *turbopause*, above which turbulent mixing no longer dominates.

You'll also notice that the slope of the "Total" line in Figure 1 changes with the composition, and is driven by the slope of each of those constituent profiles. Each species has its own "scale height" H ; in the homosphere, dominated by N₂ and O₂, the scale height is about 8 km; in the oxygen-dominated part of the atmosphere, it is about 40–60 km, and above 500 km, the scale height can reach well over 100 km. The exponential atmospheric density profile follows directly from the ideal gas law, and the scale height is given by

$$H = \frac{kT}{mg} \quad (2)$$

¹Technically, the scale height is defined as the exponential scale factor for the drop in *pressure*, not density. As the atmospheric temperature is not constant, as seen in the next section, there is some error in using it for density, but it changes the density scale height by only a few km.

²The energy of the absorbed photon is enough to *dissociate* the diatomic molecules, separating their two atoms.

where m is the molecular mass of the atmosphere. So, it is natural that in a region of the atmosphere where the molecular mass is smaller, the scale height will be larger, and thus the density profile will decrease more slowly with altitude.

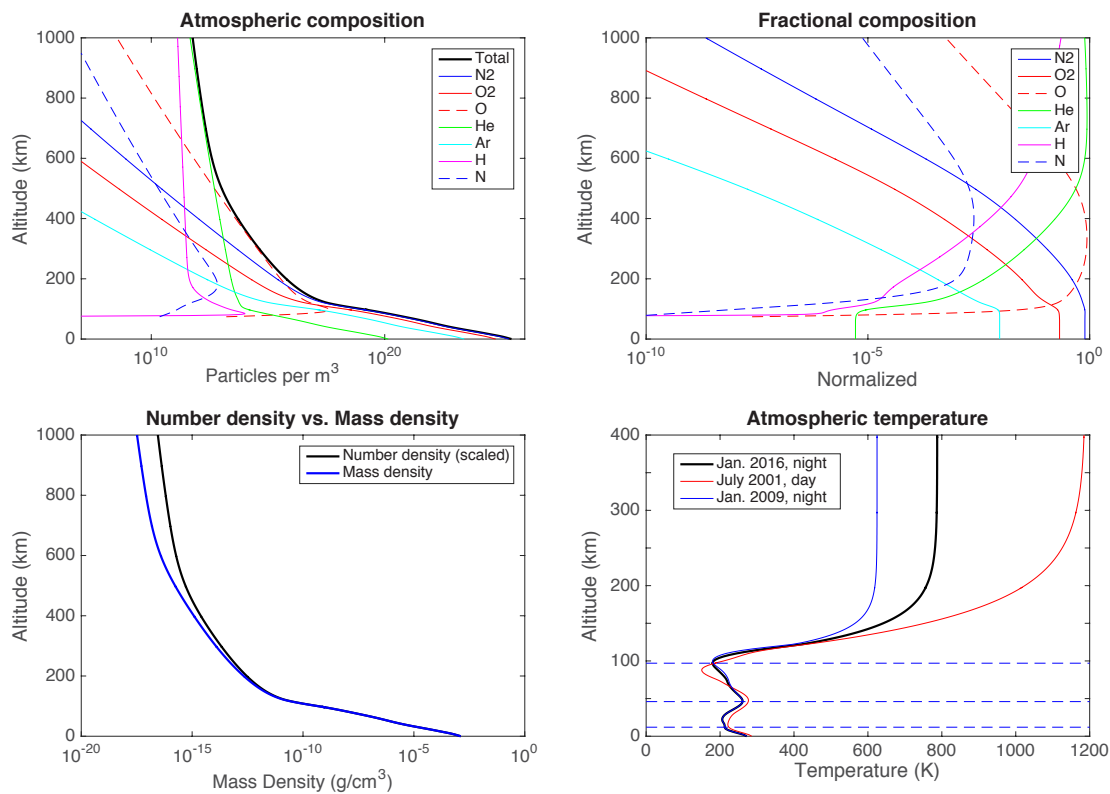


Figure 1: Atmospheric Profiles of (top left) composition, (top right) composition normalized to the total density, (bottom left) total mass density compared to number density, and (bottom right) temperature, for three different times.

3.2 Regions of the Atmosphere, by Temperature

The atmosphere can also be divided into regions by temperature. The bottom-right plot of Figure 1 shows a typical atmospheric temperature profile. The lowest region, the troposphere, extends from the ground to about 12–15 km altitude (a few km above the height of Everest); in the troposphere the temperature decreases with altitude until it reaches a minimum at the tropopause. The troposphere contains about 75% of the Earth's atmosphere by mass and contains the vast majority of the atmosphere's weather. The temperature in the troposphere decreases approximately linearly with altitude, at a *lapse rate* of about 6.5 K/km.

From the tropopause, the temperature increases with altitude in the stratosphere, until it reaches the stratopause, and a temperature maximum, at about 50 km. This increase in temperature is caused by absorption of solar UV radiation; as the top of the stratosphere absorbs more UV radiation, there is less available to heat the lower altitudes. The ozone layer resides in the stratosphere between 20–30 km. One interesting feature of the stratosphere is that the increase in temperature with altitude makes the atmosphere here dynamically stable, with very little convection and therefore little turbulence. While most aircraft fly at 10–12 km altitude (in the troposphere), the Concorde flew at 16 km altitude, in the stratosphere, and was therefore subjected to very little turbulence.

The temperature decreases again above the stratosphere, in the mesosphere, until it reaches its coldest temperature (180–200 K) at the mesopause, near 90 km altitude. The mesosphere is home to atmospheric

tides, gravity waves, layers of metal ions and neutrals left by meteors, and a complex ion chemistry.

The temperature then increases sharply above the mesopause in the thermosphere, where it is also quite “hot”, reaching temperatures of 500–1500 K, with large variation with the solar cycle. The top of the thermosphere is typically taken to be 400–600 km; above this point (the thermopause) the atmosphere is referred to as the exosphere. The delineation between the thermosphere and exosphere is primarily in *collisions*. In the exosphere, the particle density is so low that collisions are rare, and the particle trajectories are ballistic.

Page 540 of Pisacane has an insightful description of the history of the Earth’s atmosphere. Of note, the Earth’s first atmosphere was likely lost to space due to the forcing from the solar wind; this was also the fate of Mars’ atmosphere, which we will discuss later in this course.

This is probably a good time to address what “hot” really means here. Recall that temperature is essentially a measure of the average kinetic energy, or the velocity, of molecules in a Maxwell-Boltzmann distribution:

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-\frac{mv^2}{2k_B T}} \quad (3)$$

where k_B is Boltzmann’s constant, m is the particle mass, and T is the temperature. Notice that $mv^2/2$ in the exponent is simply the kinetic energy. As temperature increases, one can see that the distribution stretches to the right, meaning there are more particles at higher velocity (or energy). Example Maxwell-Boltzmann distributions are shown in Figure 2 for electrons and for atomic oxygen. The only difference between the two plots is the mass of the particle; it is clear that at the same temperature, electrons move much faster. As the temperature increases, the distribution is stretched to higher velocities and spread over a wider range of velocities.

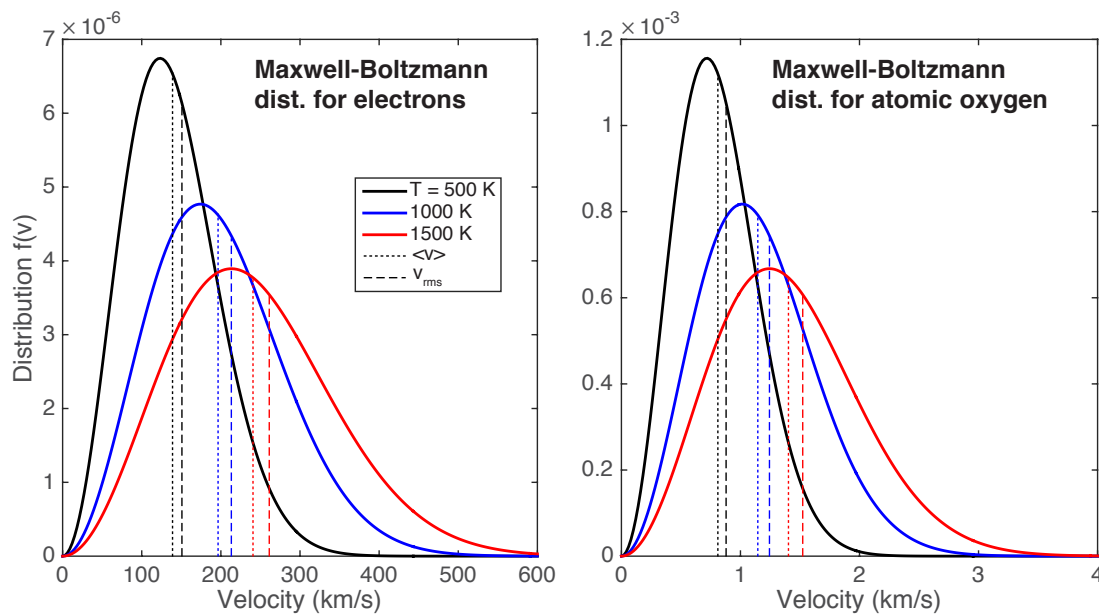


Figure 2: Maxwell-Boltzmann distributions.

The Maxwell-Boltzmann distribution is often characterized by the *mean velocity* $\langle v \rangle$ or the RMS velocity

v_{rms} , given by

$$\langle v \rangle = \sqrt{\frac{8k_B T}{\pi m}} \quad (4)$$

$$v_{\text{rms}} = \sqrt{\langle v^2 \rangle} = \sqrt{\frac{3k_B T}{m}} = 1.085 \langle v \rangle \quad (5)$$

The mean and RMS velocities are shown for the distributions in Figure 2. The mean and RMS velocities are always a bit higher than the “peak” in the velocity distribution.

On the other hand, “heat” implies a total quantity of energy, carried by a large number of molecules in this temperature-defined velocity or energy distribution. But in the thermosphere, the gas density is extremely low; even at 100 km the number density is only 10^{19} molecules/m³, some seven orders of magnitude lower than the density at the surface. So while the gas is “hot”, it doesn’t carry a lot of “heat”, and if we could step outside at 100 km altitude, it would feel extremely cold.

The transition altitudes and temperatures shown in Figure 1 are representative, but certainly not constant. The temperature profile and the altitudes of the transitions vary with latitude, solar zenith angle (SZA), and time – the solar cycle in particular has a dramatic effect on the thermosphere. This region of the atmosphere gets “hot” primarily due to absorption of solar UV radiation, which varies dramatically over the solar cycle. Figure 1 shows three different profiles taken at different times to give you an sense of the variation. The thermosphere is at its hottest in the daytime during the peak of the solar cycle (2009), and at its coldest at night during solar minimum (2001).

3.2.1 Heating and Cooling; Energy Sources and Sinks

Though less than 1.5% of the total solar irradiance, wavelengths < 300 nm are the primary source of atmospheric heating from 15–500 km. Important to heating are the processes of excitation, dissociation, and ionization (and relaxation, association, recombination). Examples of these processes follow:

- Excitation: $\text{N}_2 + \gamma \rightarrow \text{N}_2^*$
- Dissociation: $\text{N}_2 + \gamma \rightarrow \text{N} + \text{N}$
- Photoionization: $\text{N}_2 + \gamma \rightarrow \text{N}_2^+ + \text{e}^-$

To maintain thermal balance, we know there must be cooling happening in the atmosphere as well. In the thermosphere one cooling process is radiative cooling by CO₂, NO, and O (infrared emission). Other layers also cool by IR radiation, but from different components. Heat conduction downwards to the mesosphere, with heat flux proportional to temperature gradient, is another cooling mechanism. The major heating (energy sources) and cooling (energy sinks) processes for each atmospheric region is listed in Figure 3.

3.2.2 Latitude and Longitude Variation

Figure 4 shows the temperature variation in latitude and longitude in the atmosphere for different altitudes.

3.3 Other Fun Stuff in the Atmosphere

There are many other interesting features in the Earth’s atmosphere. Meteors create metal layers and noctilucent clouds shine at night. Here are some of these phenomena:

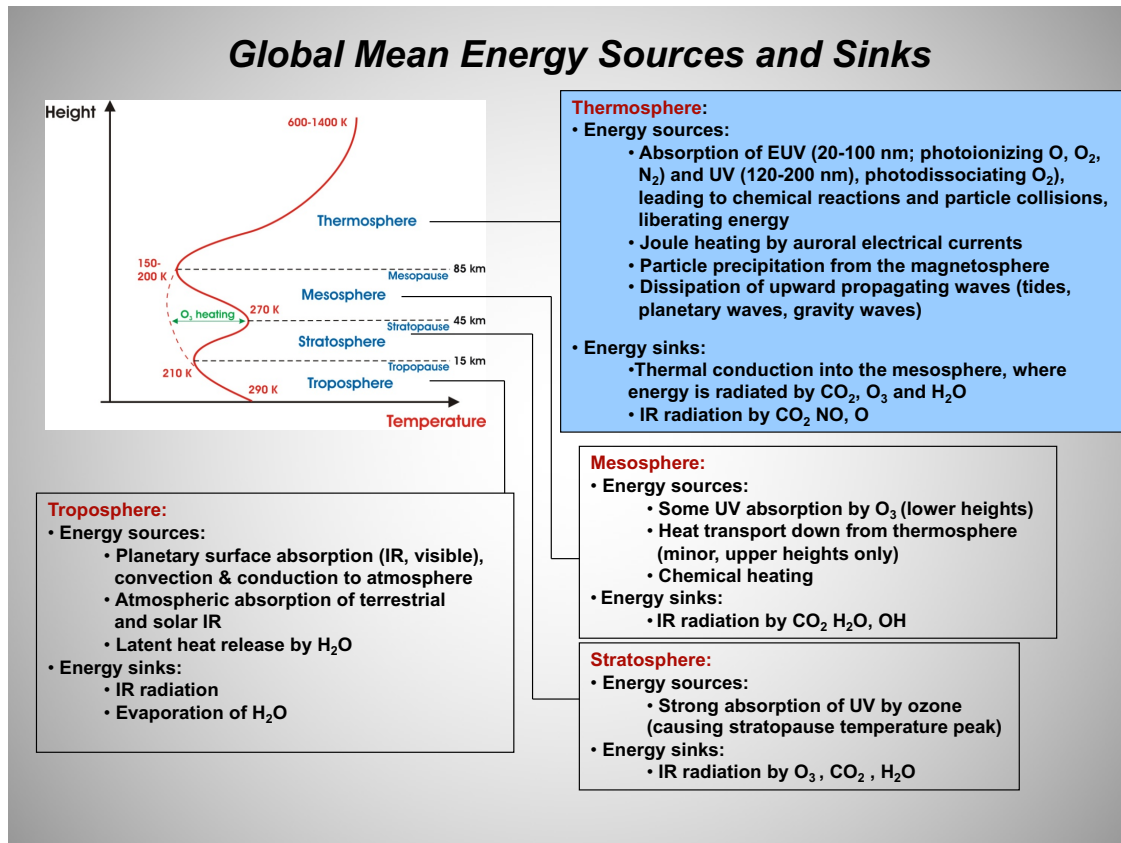


Figure 3: Sources and sinks in the atmosphere by region; courtesy D. Knipp.

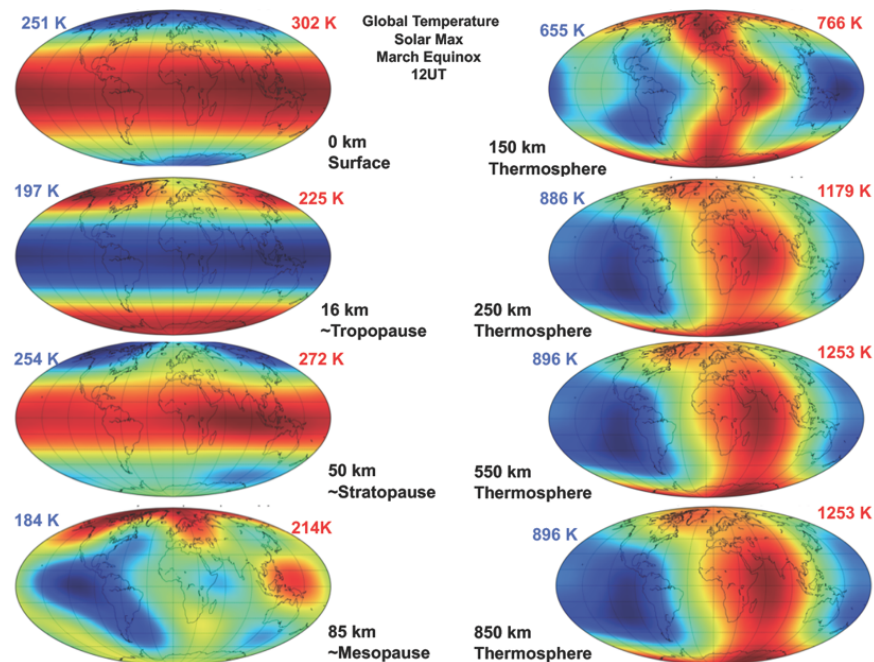


Figure 4: Temperature variation in the atmosphere with latitude and longitude for multiple altitudes.

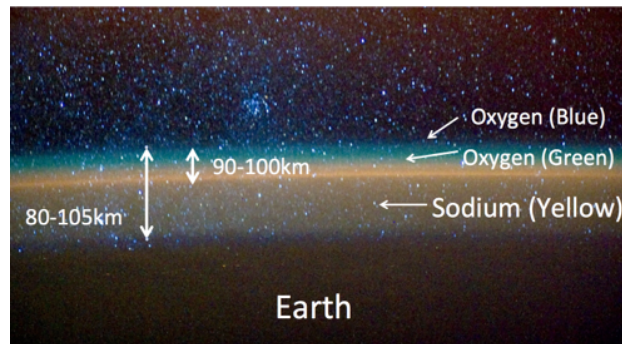


Figure 5: The sodium layer in the Earth's atmosphere is visible from space with sensitive cameras.



Figure 6: Noctilucent clouds near the mesopause. Kuresoo bog, Soomaa National Park, Estonia by Martin Koitmaa.

Metal Layers Metal Layers are created due to meteor ablation (i.e., they melt), which occurs when they enter the atmosphere. The materials, which include Ca, Na, Fe, K, and Mg, deposit into the atmosphere. With LIDAR measurements, these metal layers allow researchers (like Prof. Xinzhaoh Chu in AES) to measure the winds and temperatures of the atmosphere. A sodium layer emits bright yellow radiation, as seen in Figure 5. We will cover the metal layers and their effects more in the MMOD module later this semester.

Noctilucent Clouds Noctilucent clouds are created by ice particles at 75-85 km altitude, normally occur only in polar regions but have been observed in lower latitude regions. They are notable for being visible at twilight, illuminated by sun over the horizon, as seen in Figure 6. The Aeronomy of Ice in the Mesosphere (AIM) mission (2007-2023) was designed specifically to study these clouds and showed that they require water vapor, dust and cold temperatures; hence, they occur at the mesopause, where the temperature is minimum and dust from meteor ablation is present.

Airglow The mesosphere and thermosphere “glow” in a number of wavelengths – including those visible to the naked eye. These emissions are due to a variety of processes:

- Recombination of atoms which were photoionized by the sun during the day
- Luminescence caused by cosmic rays striking the upper atmosphere
- Chemiluminescence caused mainly by oxygen and nitrogen reacting with hydroxyl (OH) ions



Figure 7: A UHD panorama shot of the VLT platform with the red shades of airglow visible overhead from ESO (European Southern Observatory).

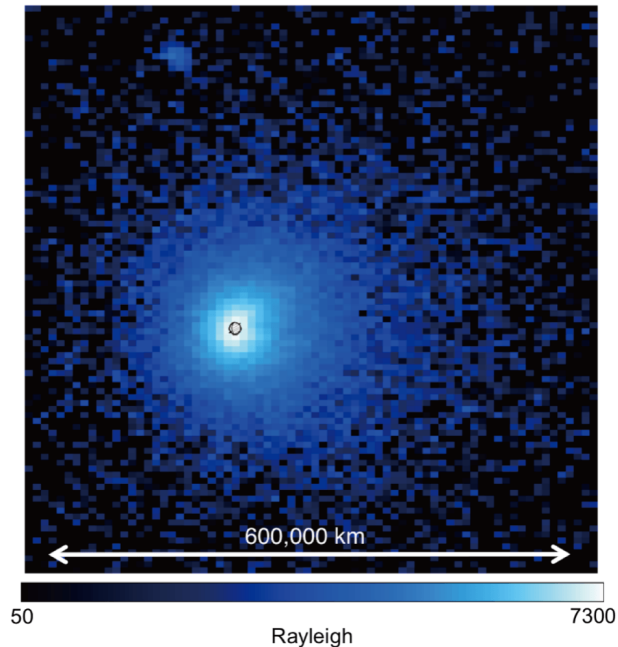


Figure 8: The hydrogen geocorona imaged by LAICA.

The airglow layers include a green layer at about 90 km altitude and a red layer at 200–400 km altitude. These layers are present day and night, but are extremely difficult to measure during the daytime. In Figure 7, the airglow is visible over a telescope platform.

Geocorona The Sun isn't the only body in the solar system with a corona. The geocorona is created from the scattering of solar Lyman- α radiation (121.4 nm) from hydrogen in the exosphere (the highest layer of the atmosphere). Due to this scattering, the Earth's atmosphere is visible out to at least 15 Earth radii ($15 R_E$). The exosphere theoretically extends to $31 R_E$, where solar radiation pressure (SRP) begins to dominate over Earth's gravity. The outer boundary of atmosphere can also be understood as the location where the SRP balances Earth's gravity. Figure 8 shows the geocorona imaged by the Lyman Alpha Imaging Camera (LAICA) onboard the Japanese PROCYON microsatellite, which launched with Hayabusa-2 in December 2014.

3.4 Atmospheric Variability and Dynamics

In this section, we will discuss the main drivers of atmospheric variability and dynamics in the stratosphere, mesosphere, and thermosphere.

3.4.1 Stratospheric Dynamics

Low in the stratosphere, there is considerable energy exchange between the stratosphere and the troposphere. The Hadley cell (a large tropical atmospheric circulation pattern) circulates and transfers tropospheric air upward, poleward, and downward. Cumulus clouds also carry mass (water vapor) into the stratosphere. From 21–35 km, there is not much hemispheric transfer, but wind reversal occurs here perennially. Above 35 km, zonal flow, driven by summer/winter heat differences, dominates heat transfer between the hemispheres.

3.4.2 Mesospheric Dynamics

Through the mesosphere, the temperature decreases with height at a rate of about 3 degrees per km due to the vibrational relaxation of the CO₂ molecules. At the start top of the mesosphere (the mesopause) the coldest atmospheric temperature is found, about 182 K (−90°C). Above 45 km, the zonal wind direction reverses (E in summer, W in winter). These winds peak at the summer and winter solstices and transition about a month after the spring and fall equinoxes, respectively.

3.4.3 Thermospheric Dynamics

The thermosphere is most interesting and important to the space environment, as this is the part of the atmosphere through which LEO spacecraft fly.

The solar energy absorbed in the thermosphere heats the ambient neutrals of the region, radiates away as airglow, and drives photodissociation of molecular oxygen into atomic oxygen. The temperature of the thermosphere is controlled by the heat sources and heat losses described in Section 3.2.1. The rates of heating and cooling driven by those processes increase with decreasing altitude (Tascione Figure 6.3, page 84) driven by the composition and density variations that affect energy transfer efficiency. Change in temperature in the thermosphere (from 500 to 1500 K) affects diffusive equilibrium of components. Generally, all particles get pushed to higher altitudes. The GUVI (Global UV Imager) instrument on the TIMED satellite measured the column density ratio of O to N₂. This is used as a proxy for neutral composition and heating of the atmosphere.

The thermosphere is affected more than other regions by the variation in the solar cycle. An increase in solar EUV increases heating. Increased heat causes the atmosphere to rise (think of the equation for scale height!) and create a higher number density in the thermosphere. It also causes increased ionization, which we will discuss in the next module on the ionosphere.

We can break up the effects that control the thermosphere by latitude. The low-latitude thermosphere is controlled by solar radiation, atmospheric tides, and ionospheric electric fields during geomagnetic storms. The mid-latitude thermosphere (above 200 km) is controlled by solar UV and EUV radiation and high-latitude phenomena during geomagnetic storms. The high-latitude thermosphere is the most variable, controlled by solar UV and EUV radiation and geomagnetic storms. During geomagnetic storms, energetic particle precipitation and changes to the ionosphere's polar electric field become significant thermosphere heat and energy sources.

Winds, tides, and gravity waves are also sources that can transfer energy to the thermosphere. Winds in the upper atmosphere are driven by at least four factors: the pressure gradient from day-to-night, the coriolis force, viscosity, and ion drag. Different effects dominate in different regions of atmosphere, both by altitude and latitude. When the winds in the thermosphere are dominated by ion drag, the wind follows ions as

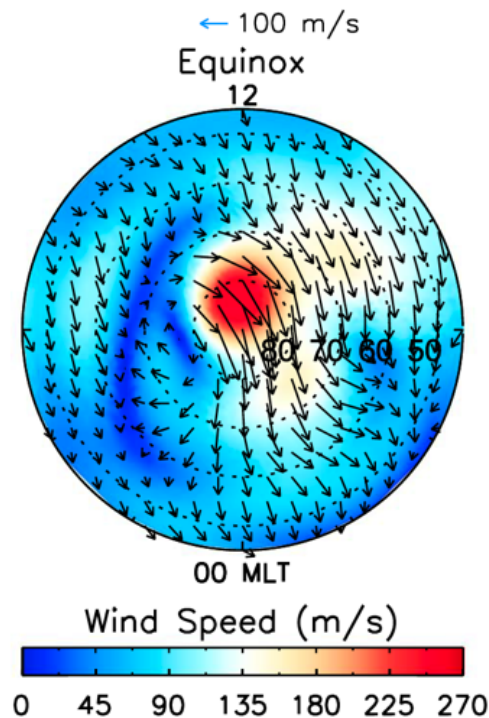


Figure 9: When the winds are ion drag dominated, the thermospheric (neutral) wind follows the ions.

illustrated in Figure 9. When thermospheric winds are pressure gradient dominated, the wind flows day to night as illustrated in Figure 10.

Tides, which are periodic oscillations in the atmosphere, are just like ocean tides but occurring in the atmosphere. They are caused by the regular day–night cycle in the Sun’s heating of the atmosphere, the gravitational pull of the Moon, non-linear interactions between tides and planetary waves, and large-scale latent heat release due to deep convection in the tropics (i.e. thunderstorms). Tides can be migrating (sun-synchronous) or non-migrating. Since the atmosphere’s density falls with altitude, tide amplitude increases with altitude exponentially.

Gravity waves in the upper atmosphere are similar to acoustic waves; in fact, they are an alternate solution to the same set of equations. Both are derived from fundamental fluid equations, but with gravity waves, the force of gravity matters. (These are not to be confused with gravitational waves!) This allows waves to propagate in the vertical direction. Gravity wave amplitude is inversely proportional to pressure, so the wave amplitude grows exponentially with altitude. Gravity waves are observed as temperature and/or density oscillations at particular altitudes. As the wave amplitude grows with altitude, at some point the low atmospheric number density cannot sustain the wave any longer and it “breaks”. This is analogous to an ocean wave breaking near the shore; the ocean depth is too shallow to sustain the wave height. A simulation of gravity waves is shown in Figure 11. Gravity waves are initiated in the troposphere; they can be triggered by thunderstorms, by winds flowing over mountain ranges, or a number of other sources.

3.5 Measuring and Modeling the Atmosphere

So why does atmospheric variability matter? For space weather, temperature changes, winds, waves, and tides all affect density and/or velocity of the atmosphere, which affect spacecraft drag, spacecraft and radio communications, and oxygen effects. We’ll talk more about these effects in Section 4. But first, let’s talk about how we measure and predict atmospheric variations.

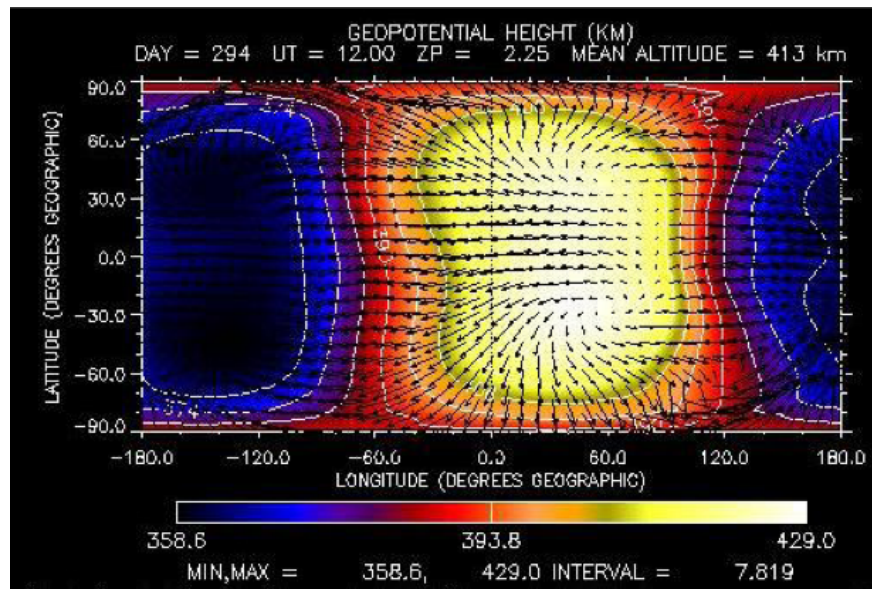


Figure 10: When the wind is pressure gradient dominated, the thermospheric wind flows day-to-night.

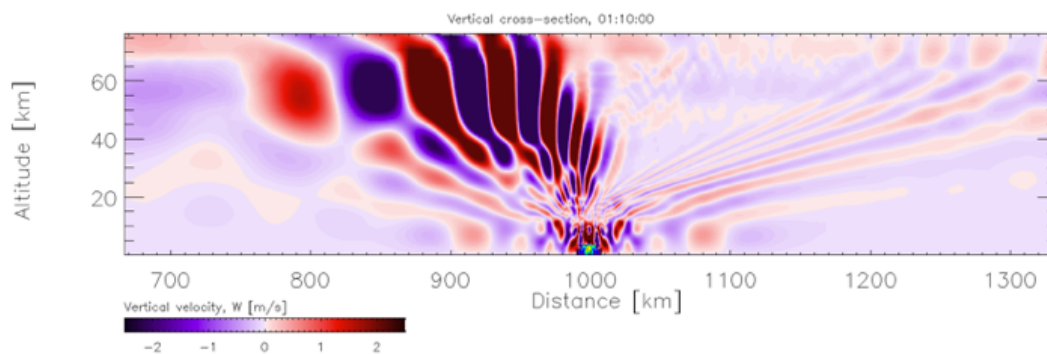


Figure 11: Simulation of gravity waves with vertical velocity shown on a plot of altitude vs. distance. Courtesy J. Snively, Embry-Riddle Aeronautical University.

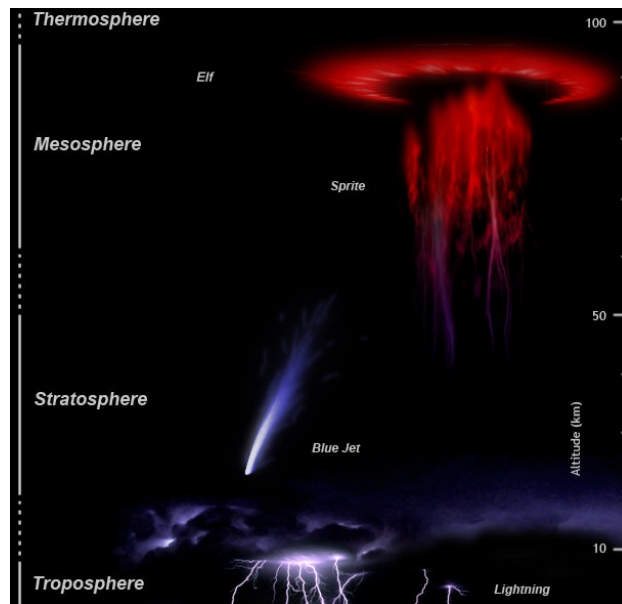


Figure 12: Lightning effects in the upper atmosphere.

There are many atmospheric models utilized to predict atmospheric conditions including MSIS, the US Standard Atmosphere, the International Standard Atmosphere, the Jacchia model, and others. We will focus on MSIS, which stands for mass spectrometer and incoherent scatter radar (named for the source of data used to create the models). The MSIS models are global, empirical models of atmosphere density, composition, and temperature. They are derived from data from mass spectrometers, radars, rocket measurements, and more. MSIS includes variations with solar activity, location, and more. There are a few versions of MSIS hosted by NASA's Community Coordinated Modeling Center (CCMC): [MSIS-E-90](#) and [NRLMSISE00](#). You will utilize an atmospheric profile produced by MSIS in the homework in this module!

3.6 Lightning Effects in the Atmosphere

Lightning couples with the atmosphere from above and below affecting electric fields and ionization. From below, lightning modifies the background electric field. At high altitudes, the perturbed E-field can exceed breakdown, triggering sprites! A lightning electromagnetic pulse (EMP) can propagate up to base of ionosphere, where it heats electrons. This causes new ionization and light emissions, including elves. In fact, roughly 10% of all cloud-to-ground lightning produces elves. Elves and sprites are pictured in Figure 12.

4 Effects

Earth's atmosphere is important for spacecraft below about 800 km; above that, the density is simply too low to have any real impact, and the space environment is dominated by the magnetized plasma of the ionosphere and magnetosphere. Hence, this discussion will focus entirely on LEO spacecraft.

4.1 Spacecraft drag

Let's quickly derive the drag equation. Consider a spacecraft with mass m (kg) and cross-section area A (m^2) flying through a gas of mass density ρ (kg/m^3). From the frame of reference of the spacecraft, the gas

is flowing across the spacecraft at a velocity v (m/s), like wind hitting the spacecraft. The dynamic pressure of this flowing air is given by

$$P_{\text{dyn}} = \frac{1}{2} \rho v^2 \quad (6)$$

Pressure is force per unit area, or $P = F/A$; so we can write the drag force as $F = PA$ or

$$F_{\text{drag}} = \frac{1}{2} \rho v^2 C_D A \quad (7)$$

Notice that we sneakily added an extra constant C_D , the drag coefficient. C_D is related to the Reynolds number and the spacecraft shape, and is not easily derived; it is usually empirically determined or approximated. Even the drag coefficient of a sphere, the simplest possible spacecraft shape, has a complicated drag coefficient as a function of Reynolds number. For spacecraft in LEO the drag coefficient is usually $C_D \simeq 1-3$.

Let's consider a realistic case. I have a 3U CubeSat ($10 \times 10 \times 30$ cm, 4 kg) flying with its 10×10 cm face in the ram direction, so $A = 0.1 \times 0.1 = 0.01$ m². At 400 km altitude (the altitude of the ISS) the atmosphere's mass density is approximately 10^{-15} g/cm³ (we'll use the MSIS models to get a more accurate value for real calculations). The drag coefficient for a typical CubeSat is often approximated to be $C_D = 2.2$ (though it can vary). From Kepler's Law we find that the orbital speed at 400 km is about 7.7 km/s.

Taking all of these numbers into account, the drag force on the spacecraft is in the range of μN (microNewtons). That seems small; perhaps drag is not such a big issue at 400 km?

Here's a simple question: does drag cause a satellite to slow down, or to speed up?

Your intuition probably tells you that a satellite will slow down, but your intuition is wrong. Recall that to first order, the orbital velocity of a satellite in circular orbit is given by

$$v = \sqrt{\frac{GM_E}{r}} \quad (8)$$

where r is the distance from the center of the Earth to the satellite. When a satellite is subjected to drag, its orbit will change and its altitude will be reduced; but if r gets smaller, v gets larger! The satellite loses kinetic energy due to drag, but it is more than made up for by the kinetic energy it gains while losing potential energy by reducing r .

Recall from Kepler's third law that in circular orbit, the orbital period P is given by

$$P^2 GM_E = 4\pi^2 a^3 \quad (9)$$

where a is the orbital radius (altitude plus Earth radius), G is the gravitational constant, and M_E is the mass of the Earth. We can combine equation (7) (writing $F = m dv/dt$) and equations (8) and (9) to determine the lifetime of a spacecraft. You'll do this as a homework problem to find that the change in period with time is given by

$$\frac{dP}{dt} = -3\pi a \rho \frac{C_D A}{m} \quad (10)$$

The term $C_D A/m$ is sometimes simply written as the inverse ballistic coefficient β^{-1} . Using the two equations above, we can see how the orbit will change with time. At a given time and location there is an atmospheric mass density ρ , which leads to a change in the orbit period $P' = P - dP$ after some time dt . With this new period P' , we can calculate a new altitude a' from equation (9). Since equation (10) has a negative sign, $P' < P$ and $a' < a$, i.e. the altitude will be lower. Now at this new, lower altitude, $\rho' > \rho$, and we will have to update the density. We can then iterate through these two equations, updating ρ at each new altitude, to see how the orbit changes with each time step.

Figure 13 shows an example of this orbit decay, plotting the altitude, period, and velocity as a function of time for a spacecraft that was launched at 400 km or 500 km altitude.

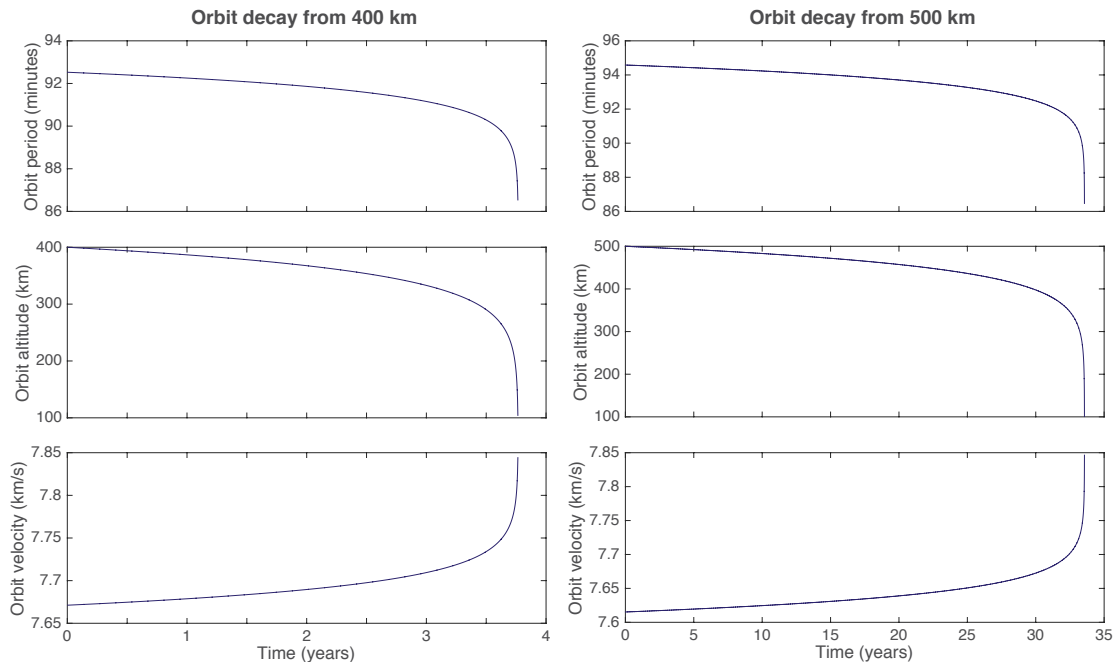


Figure 13: Decay of spacecraft orbits from 400 or 500 km altitude.

4.1.1 Density and Solar Activity

Solar variation causes the atmospheric density to change. During solar maximum, the atomic oxygen density is much higher than during solar minimum. Atmospheric tides and gravity waves create fluctuations in the density.

Let's assume we know the atmospheric mass density to within an uncertainty of 50%. That mass density uncertainty leads to uncertainty in the drag force, which leads to uncertainty in a , P , and velocity, since they are all tied together by the equations above. If there is uncertainty in the velocity, this translates into an accumulating position uncertainty over time. Consider that a typical spacecraft will fly over a mid-latitude ground station (such as Boulder) about twice per day; then the ground station pass will occur later or earlier than our basic orbital mechanics will predict. Ground station operators need to take this into account, make regular observations of spacecraft, and constantly update their positions to zero out this growing position uncertainty. We'll explore this issue further in a homework problem below.

4.1.2 The Ballistic Coefficient

One factor we have glossed over is the ballistic coefficient. The ram-directed cross section area A and the mass m ought to be well known, but the drag coefficient C_D can be difficult to determine. The drag coefficient varies for different shapes and materials, as well as the flow conditions of the atmosphere.

4.2 Absorption of Solar Radiation

The atmosphere absorbs a good chunk of the total solar irradiance (TSI). Comparing the TSI just outside the atmosphere to at the Earth's surface, we see a reduction from 1361 W/m^2 to 1000 W/m^2 . The effect of atmospheric absorption then is important to know this for solar panels on the ground. The atmosphere is especially good at absorbing EUV due to ozone (O_3) and other molecules. Without ozone, we all get our skin burned and are likely to get skin cancer! The ozone layer is critical to life on Earth, despite consisting

of only 10 ppm. If you condense all the ozone into a thin layer at 1 atm (sea level pressure), this layer would be only 3 mm thick. Ozone can be destroyed by NO, N₂O, OH, Cl, and Br, the latter two of which are products of CFCs and BFCs.

4.3 Atomic Oxygen effects

Recall from Figure 1 that at LEO spacecraft altitudes of 300–1000 km, the atmosphere is dominated by atomic oxygen (O) and/or helium (He). As it turns out, atomic oxygen is a pretty nasty gas for spacecraft to have to live in. Oxygen is highly reactive, even here on the surface where many metals oxidize. In space, exposure to oxygen leads to oxidative erosion, as the spacecraft surfaces react with atomic oxygen and material is removed.

In addition to erosion, atomic oxygen can result in accommodation, where the oxygen molecules “stick” to the surfaces of the spacecraft. This accommodation has two secondary effects: first, it can change the drag coefficient, thus contributing to the issues of spacecraft drag and making orbit prediction difficult; and second, it can change the surfaces’ emissivity and absorptivity, affecting the thermal balance of the spacecraft.

4.3.1 Oxygen Attack

Atomic oxygen is the dominant species in LEO. O leads to the erosion of spacecraft materials, with the amount of damage depending on the material. The mass loss resulting from the erosion of some flux of O is given by:

$$dm = \rho_t RE \phi dA dt \quad (11)$$

where ρ_t is the density of the material (g cm^{-3}), RE is the reaction efficiency (cm^3 per O atom), ϕ is the O flux (O atoms/ cm^2/sec), dA is the surface area, and dt is the time the flux acts. This can be rewritten as:

$$\frac{dx}{dt} = RE \phi \quad (12)$$

where $\frac{dx}{dt}$ is the rate of change of the material thickness. O induced erosion is purely chemical, and defect in protective coatings (caused from manufacturing or MMOD impacts) will allow O to penetrate to and degrade underlying surfaces (thus, Oxygen attack).

Figure 14 shows typical reaction efficiencies (RE) for common materials interacting with atomic oxygen.

4.3.2 Air-breathing Thruster

Atomic oxygen isn’t all bad! An air-breathing thruster is designed so that a “scoop” captures the atomic-oxygen-dominated air as it orbits. Then, an ionization chamber ionizes the air, and an accelerator grid accelerates the ions to high speeds (tens of km/s) as exhaust. Currently, air-breathing thruster is a hot research topic because of its potential application to prolong the lifetime of VLEO (Very Low Earth Orbit) spacecraft. In the homework, you will determine how much thrust this mechanism would need to create to balance the force of drag, and what velocity of the atomic oxygen exhaust would be required.

4.4 Sputtering

Sputtering refers to the process by which impacts from neutral atoms or molecules knock material from a spacecraft surface. This can only occur if the molecule has a higher energy than the chemical bonds on the

Table 3.7
Atomic Oxygen Reaction Efficiency

Material	Reaction Efficiency ($\times 10^{-24}$ cm ³ /atom)	
	Range	Best Value
Aluminum	-	0.00
Carbon	0.9–1.7	-
Epoxy	1.7–2.5	-
Fluoropolymers		
- FEP Kapton	-	0.03
- Kapton F	-	<0.05
- Teflon, FEP	-	<0.05
- Teflon	0.03–0.50	-
Gold	-	0.0
Indium Tin Oxide	-	0.002
Mylar	1.5–3.9	-
Paint		
- S13GLO	-	0.0
- YB71	-	0.0
- Z276	-	0.85
- Z302	-	4.50
- Z306	-	0.85
- Z853	-	0.75
Polyimide		
- Kapton	1.4–2.5	-
- Kapton H	-	3.04
Silicones		
- RTV560	-	0.443
- RTV670	-	0.0
Silver	-	10.5
Tedlar		
- Clear	1.3–3.2	-
- White	0.05–0.6	-

Figure 14: Reaction efficiencies for different materials interacting with atomic oxygen; from *Tribble*.

Table 3.5
Sputtering Thresholds

Target Material	Bombarding Gas Threshold (eV)					
	O	O ₂	N ₂	Ar	He	H
Ag	12	14	13	17	25	83
Al	23	29	27	31	14	28
Au	19	15	15	15	53	192
C	65	82	79	88	40	36
Cu	15	22	21	24	20	60
Fe	20	28	27	31	23	66
Ni	20	29	27	31	24	72
Si	31	39	37	42	18	40

Source: From *Planet. Space Sci.*, 36, R. R. Laher and L. R. Megill, "Ablation of Materials in the Low Earth Orbital Environment," pp. 1497–1508, © 1988. (Reprinted with kind permission of Pergamon Press Ltd, Headington Hill Hall, Oxford OX3 0BW, UK.)

Figure 15: Sputtering thresholds for different bombarding materials incident on target materials; from *Tribble*.

surface of the spacecraft. Sputtering degrades surface coatings and solar cells, posing a danger to spacecraft. Some sputtering thresholds (how energetic a molecule must be to damage a target material) are shown in Figure 15, taken from Tribble.

You'll compare these thresholds to the actual impact energy of atmospheric atoms and molecules in one of the homework problems.

4.5 Vacuum Effects

When we talk about vacuum effects, we are really talking about lack-of-atmosphere effects. There are a variety of effects that happen in vacuum precisely due to the absence of atmosphere. Here, we list a few:

- Outgassing – A spacecraft itself can be a source of contamination, releasing volatile chemicals during launch or in orbit. Chemicals within a material can migrate towards the surface and be released over time and impact other surfaces.
- Solar UV degradation – UV effects are covered in *Module 1: The Sun*.
- Pressure differentials – Spacecraft must be designed to handle pressure changes throughout orbits. Consider a manned spacecraft pressurized to 101 kPa. It would exert 101 kN of force on a 1 m² structure. The spacecraft must be able to handle this force. Sudden depressurization is also a concern for manned missions.
- Particulate contamination – Since particulates on surfaces will absorb or reflect light, they can have a large impact on optical sensors, creating false signals, affecting SNR, and limiting off-axis rejection. These particulates may originate from outgassing (above), venting of pressurized cavities, or dust in the space environment.
- Solar array degradation – Contaminants can build up on solar arrays, degrading their performance.
- Optical effects – Contaminants on lenses, mirrors, or focal planes will degrade the SNR of an optical detector, as the contaminants will absorb light from the target.

5 Missions

5.1 CHAMP

The [CHAMP spacecraft](#) was designed to measure the LEO space environment, including the mass density and composition. In lecture I'll present some measurements and insights from CHAMP.

5.2 Planet Labs

[Planet](#) has a constellation of >100 CubeSats in low-Earth orbit for Earth imaging. These 3U CubeSats don't have any onboard propulsion, but the company wants them to maintain a particular constellation configuration (i.e. spacing) for their mission lifetime. They accomplish orbit modifications through *differential drag*. By changing the area facing the ram direction, they can change the drag force on the spacecraft, and move the spacecraft into slightly different orbits with respect to each other. There's a paper posted on Canvas that describes this in detail.

6 Movie: Spaceballs

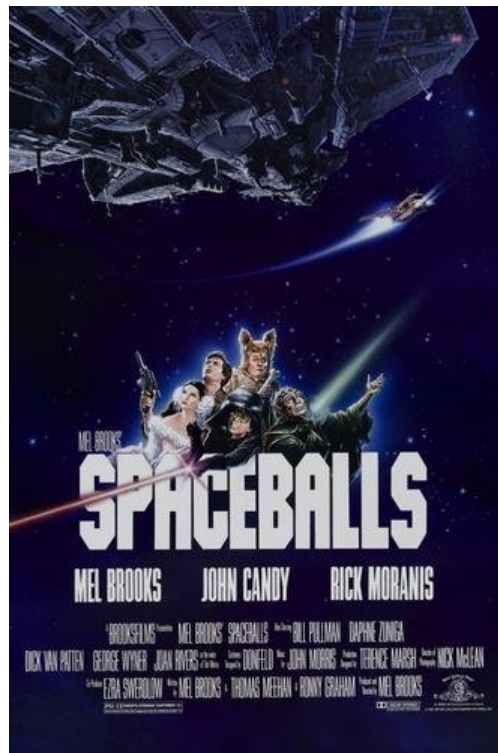


Figure 16: Spaceballs (1987).

Synopsis from Fandango:

In a distant galaxy, planet Spaceball has depleted its air supply, leaving its citizens reliant on a product called "Perri-Air." In desperation, Spaceball's leader President Skroob (Mel Brooks) orders the evil Dark Helmet (Rick Moranis) to kidnap Princess Vespa (Daphne Zuniga) of oxygen-rich Druidia and hold her hostage in exchange for air. But help arrives for the Princess in the

form of renegade space pilot Lone Starr (Bill Pullman) and his half-man, half-dog partner, Barf (John Candy).

Think about [this scene](#) when you work out Problem 1 below.

7 Problems

1. **(20) Spaceballs!** Check out the movie clip linked above (also available on Canvas).
 - i) What is the total mass of the Earth's atmosphere? Use the MSIS profile posted on Canvas (MSISatmosphere1000.m).
 - ii) If you crammed all of this atmosphere into a vacuum bag $10 \times 10 \times 100$ km in volume, at a temperature of 300 K, what would be the pressure in the bag? Assuming the molar mass for the air: 0.029 kg/mol.
2. **(10) Atmosphere Scale Heights:** Calculate the approximate scale heights for "air" (N_2/O_2 mixture at sea level), atomic oxygen, helium, and hydrogen, assuming a constant temperature of 300 K. Are these scale heights consistent with the profiles in Figure 1? Compare quantitatively, by calculating the scale height using the MSIS profiles again, at a few altitudes in the atmosphere appropriate to these three constituents. Explain qualitatively why the scale heights from these two methods differ.
3. **(10) Atmosphere Variation:**
 - i) Use the [NRLMSISE-00 model online](#) to generate your own profiles of mass density versus altitude for solar minimum and solar maximum, up to 1000 km altitude. I suggest using 2001 for solar maximum, and 2009 for solar minimum; use January 1, 1200 UT, and 0 latitude and 0 longitude. Plot both profiles on a single axis and submit that figure with your assignment.
 - ii) What is the variation between solar maximum and solar minimum at 600 km altitude? What are the implications of this variation for spacecraft operations? (We're just looking for qualitative comments.)
 - iii) What is the dominant atmospheric constituent at 700 km during solar maximum? What about solar minimum?
4. **(10) Oxygen Impact:** If the temperature in the thermosphere is 1000 K,
 - i) What is the characteristic velocity of an oxygen atom (O)?
 - ii) How much energy (in eV) is imparted to a spacecraft in LEO (say, 600 km altitude) for each collision with an O atom?
 - iii) What fraction of the atomic oxygen population has sufficient energy to cause sputtering on Aluminum surfaces? (See Figure 15 above. Consider just the 1D velocities parallel to the spacecraft ram direction.)
5. **(20) Orbit Decay:**
 - i) Derive equation (10) that relates the change in orbit period to the mass density of the atmosphere.
 - ii) The satellite (a 3U CubeSat) in Figure 17 with $C_D = 2.2$ is in orbit at 400 km altitude. If the satellite is flying with its ram direction in the $+x$ direction shown, how long until it deorbits (reaching at 120 km)? Again, use the MSIS profile posted on Canvas for the mass density of the atmosphere. You'll need to numerically (i.e. iteratively) solve the equation for dP/dt . Plot the satellite altitude versus time.

- iii) Repeat for a starting altitude of 500 km.
- iv) Repeat for a starting altitude of 400 km again, but now with the spacecraft flying in the $+z$ direction.

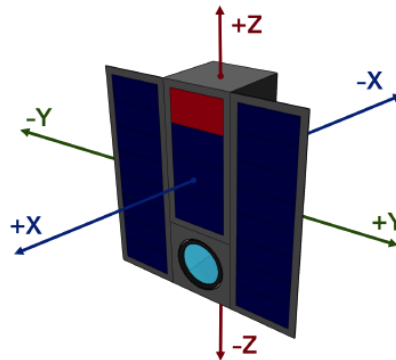


Figure 17: CubeSat for problem 5. The CubeSat is $10 \times 10 \times 30$ cm, and the two deployed solar panels are 10×30 cm each.

6. **(10) Atmosphere Uncertainty and Neutral Winds:** Let's add some uncertainties in our calculations for the problem above. Consider the first case from above, where our spacecraft is at 400 km altitude, flying in the $+x$ spacecraft direction.
- i) Now assume there is an atmosphere density uncertainty of 10%. Repeat the calculation from the previous problem. Plot the altitude vs. time, showing both cases (this one and Problem 5) on the same plot. What is the difference in deorbit time?
 - ii) Given that the orbit around the Earth is roughly 90 minutes (give or take) at any LEO altitude, what are the implications for where on Earth this spacecraft will re-enter the atmosphere? (i.e. can you predict where it will re-enter? What's the uncertainty in the deorbit location?)
 - iii) BONUS (5 points): Repeat the calculation, but assume a constant neutral wind of 100 m/s directed towards the spacecraft ram face (i.e., a headwind). *[This is not trivial! You have to re-derive the equation that you are iterating; why?? If you don't get to this part, don't worry about it.]*
7. **(10) Air-breathing Thruster:** Imagine a spacecraft at 400 km that lets in the neutral atmosphere through an inlet, ionizes it, and accelerates it out the back (an example is shown in Figure 18). The spacecraft has $C_D = 2.2$ and $A = 100 \text{ cm}^2$.
- i) What thrust is required to balance the drag force?
 - ii) What is the required exhaust velocity of the atomic oxygen to achieve this thrust? (hint: $\Delta m / \Delta t$, the exhaust mass rate, must be the same as the mass collection rate.)

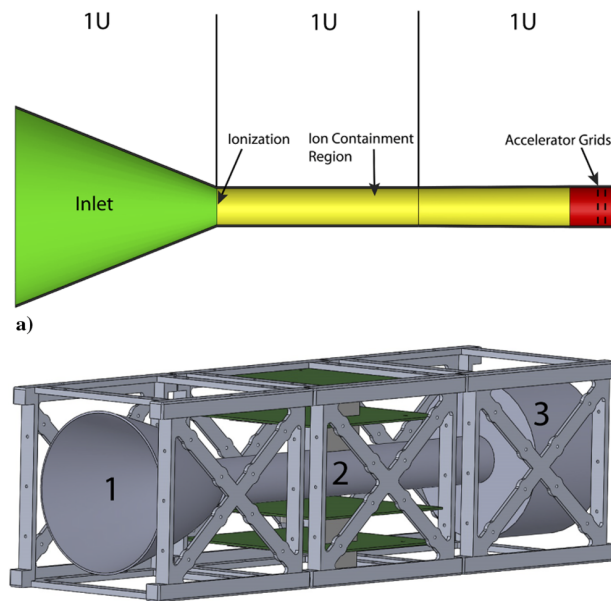


Figure 18: Concept design for a CubeSat with an “air-breathing thruster”. 1) inlet; 2) ionization chamber; 3) ion accelerator. From Jackson and Marshall (2017).

8. **(10) Missions:** How do we measure the number density, and/or the composition, of the thermosphere? Research one of the following methods. Describe how the method works and describe an instrument on a past or current mission (or on the ground) that makes these measurements.

- Space-based Ion and Neutral Mass Spectrometer (INMS)
- Ground-based incoherent scatter radar (ISR)
- Space-based accelerometer measurements (e.g., CHAMP)
- Another method that you discover in your literature search